

Deployment Corpus Verification

Text-extraction and chunk-integrity QA across all deployment countries

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Overview

This notebook verifies the text-extraction quality of every country in the deployment pipeline (currently Malaysia; future Indonesia, Thailand, Philippines, Vietnam). It runs on `country_body` and `country_chunks` after they are built, before C1 deployment.

The US precedent is `verify_body.qmd`. Six of its seven tests port directly: URL resolution, boundary documents, temporal coverage, text quality, anomaly detection, and a new chunk-integrity test that the US notebook does not run. The seventh US test, “Known Act Validation”, is omitted because deployment countries have no labeled passages. Anchor-event detection (1997 Asian Financial Crisis, 2015 GST, 2020 COVID packages) is reserved for the C1-output deployment evaluation notebook, not here.

Corpus snapshot

Table 1: Corpus snapshot per country

Coun- try	Doc- uments	Extracted Pages	Year range	Doc-Sources ument bod- ies	OCR docs	OCR Chunks pages			
malaysia	162	107	20,945	1980-2022	6	10	48	5,010	3,033

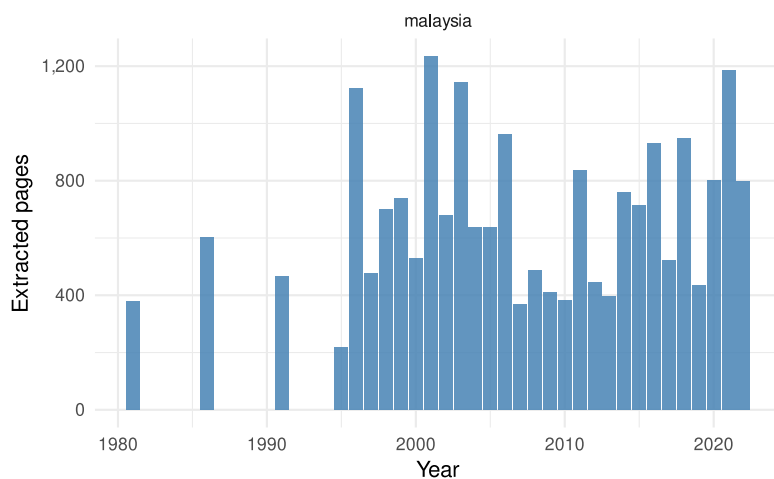


Figure 1: Extracted pages per year, by country

Test (i): URL resolution and extraction success

For each country, the share of documents in `country_urls` that produced extracted text in `country_body`. Documents with `access_status == "manual_pending"` are reported separately as known acquisition gaps, not extraction failures.

Table 2: Test (i) URL resolution: outcomes per country

Coun- try	Expected	Pend- ing	Re- solv- able	Extracted	Failed	Suc- cess rate	Sta- tus
malaysia	154	55	99	99	0	100.0%	PASS

Table 3: Test (i) URL resolution: by country and document body

Country	Body	Ex- pected	Pending	Ex- tracted	Failed
malaysia	Bank Negara Malaysia Annual Report	43	17	26	0
malaysia	Budget Speech / Ucapan Bajet	43	19	24	0
malaysia	Crisis booklet / fiscal stimulus package	8	0	8	0
malaysia	Eco- nomic Report / Tinjauan Ekonomi	43	15	28	0
malaysia	Five- Year Malaysia Plan / Mid- Term Re- view	17	4	13	0

Table 4: Test (i) Failed extractions (excluding pending)

No failed extractions among resolvable documents.

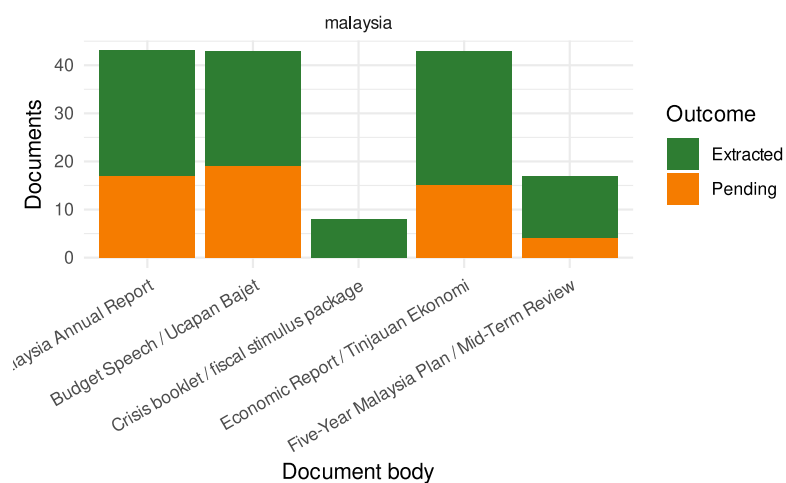


Figure 2: Document outcomes by country and body

Test (i) is the headline gate. Pending rows reflect the manual-acquisition workflow and are tracked but excluded from the success-rate denominator. A FAIL here points to URL resolution or PDF-extraction problems and must be fixed before C1 deployment runs.

Test (ii): Boundary documents

For every country × source × body combination, the earliest and latest documents with extracted text must be at least 10 pages long. This catches systematic extraction failure at the edges of the corpus where document formats are most likely to drift.

Coun- try	Source	Body	Year	Bound- ary	Pages	OCR
malaysia	helvinsjawan.mof.gov.my	Budget Speech / Ucapan Bajet	1999	Earliest	32	FALSE
malaysia	helvinsjawan.mof.gov.my	Budget Speech / Ucapan Bajet	2022	Latest	168	TRUE
malaysia	helvinsjawan.mof.gov.my	Economic Report / Tinjauan Ekonomi	2014	Earliest	219	TRUE
malaysia	helvinsjawan.mof.gov.my	Economic Report / Tinjauan Ekonomi	2022	Latest	204	FALSE
malaysiabnm.gov.my		Bank Negara Malaysia Annual Report	1997	Earliest	253	FALSE
malaysiabnm.gov.my		Bank Negara Malaysia Annual Report	2022	Latest	211	TRUE
malaysia	scholar.library.yale.edu	Crisis booklet / fiscal stimulus package	1998	Earliest	197	TRUE
malaysia	scholar.library.yale.edu	Crisis booklet / fiscal	1998	Latest	197	TRUE
malaysia	theftfma.org.ph	GFRIS Yearly PHANERMA report Re-Re-published page from	2000	Earliest	300	FALSE

**Malaysia — Budget Speech / Ucapan Bajet, Earliest 1999
(belanjawan.mof.gov.my)**

undertaken:
generating growth through fiscal stimulus;
reviving business activities; and
improving the efficiency of the public sector and co-
operation with the private sector.

a.
Generating Growth Through Fiscal Stimulus

76.
Fiscal stimulus is necessary to speed up economic
recovery. However, the
Government will ensure that the nation registers a
surplus or at least a balance in its
current account position. This is an important
principle in prudent financial management
to ensure that Government revenue is at least
sufficient to finance operating expenditure.
Having taken into account the need to finance
development expenditure, the deficit in
the Federal Government's financial position will be
limited to 6 percent of GNP. This
deficit will be financed from borrowings from both
the do...

**Malaysia — Economic Report / Tinjauan Ekonomi, Earliest
2014 (belanjawan.mof.gov.my)**

Economic Performance and
Prospects
economic report 2013/2014
97
TABLE 3.8
Palm Oil Areas, Yield and Production
2012 – 2013
Change
(%)
2012
2013
2012
2013
Planted areas
('000 hectares)

5,077
5,177
1.5
1.9
Matured areas
('000 hectares)
4,353
4,442
1.5
2.0
Production
('000 tonnes)
Crude palm oil
18,785
19,414
-0.7
3.3
Crude Palm kernel oil
2,164
2,200
0.9
1.7
Yield (tonnes/
hectare)2
18.9
19.9
-5.0
5.3
1 Estimate.
2 Fresh fruit bunches yield.
Note: Total may not add up due to rounding.
Source: Ministry of Plantation Industries and
Commodities, and
Ministry of Finance, Malaysia.
TABLE 3.9
Rubber Areas, Yield and Production
2012 – 2013
Change
(%)
2012
20131
2012
20131

```

Total areas
('000 hectares)
1,042
1,050
0.2
0.8
    Smallholdings
977
986
2.3
0.9
    Estates
64
64
0....

```

Malaysia — Bank Negara Malaysia Annual Report, Earliest 1997 (bnm.gov.my)

```

Conclusion
In a globalised world market, efforts to
enhance international competitiveness through
incentives, lowering of tax rates and other
orthodox measures are clearly no longer
adequate. In addition, the globalisation process
has also opened new avenues through which
risks can spread rapidly, posing potential
systemic damage to the financial system and the
economy. While the recent depreciation of the
ringgit, to some extent, will help to strengthen the
competitiveness of exports, the nation should
not rely solely on the exchange rate to sustain
long-term competitiveness. Improving the,
competitive position in the longer term is a
structural issue to be attained through the
development of new approaches, strategies and
products.

```

A short boundary document does not always indicate failure (a 2020 crisis booklet may legitimately be 8 pages), but every flagged row warrants a glance at the URL and a re-pull if the page count is implausibly low.

Test (iii): Temporal and source coverage

The expected document grid is `country_urls` itself: every row in the URL manifest is a document the pipeline expects to extract. This is cleaner than the US notebook's hardcoded coverage grid because each country's URL fetcher already encodes its own coverage rules.

Table 6: Test (iii) Coverage rate per country

Country	Total	Present	Pending	Missing	Coverage rate	Status
malaysia	154	99	55	0	100.0%	PASS

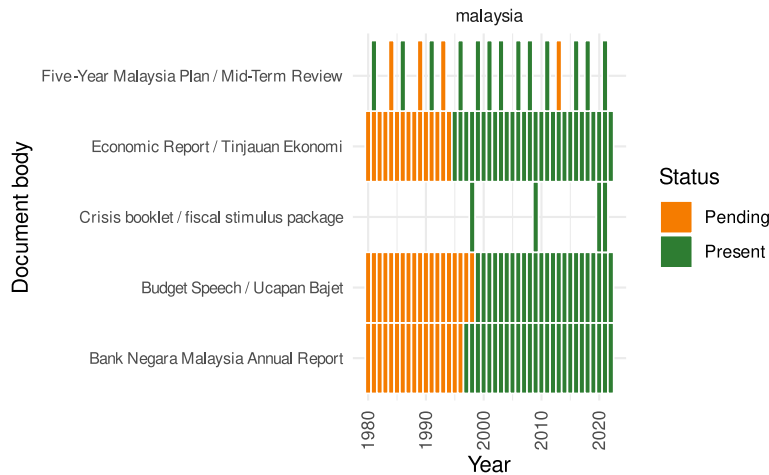


Figure 3: Coverage heatmap by country, body, and year

Table 7: Test (iii) Missing expected documents

No missing documents among non-pending expectations.

A WARN here is acceptable if the missing rows cluster in a single series with known access issues; a FAIL means the coverage assumption built into the URL fetcher is broken for some series. Either way, the missing-rows table is the actionable artifact for the acquisition workflow.

Test (iv): Text quality

Page-level metrics computed on a sample of documents per country, with fiscal-vocabulary detection adapted per country so the test does not silently fail on non-US documents (US uses \$; Malaysia uses RM and `ringgit`).

Table 8: Test (iv) Text quality summary per country

Country	Pages sampled	Suspicious page rate	Fiscal vocab rate	Status
malaysia	250	7.6%	81.2%	WARN

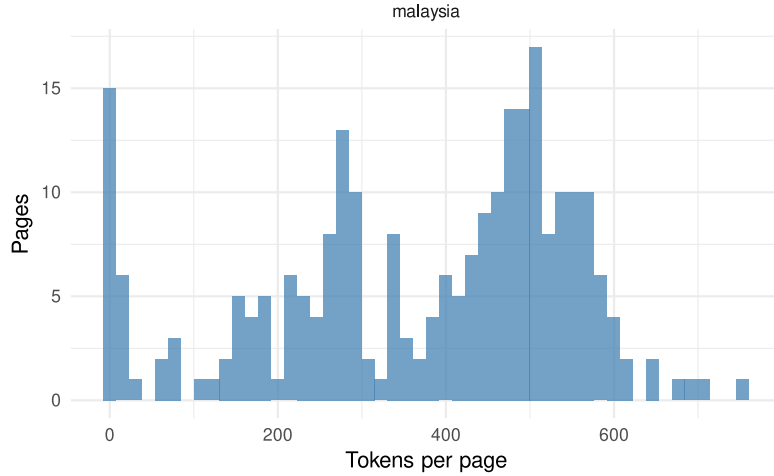


Figure 4: Token distribution per page across the sampled documents

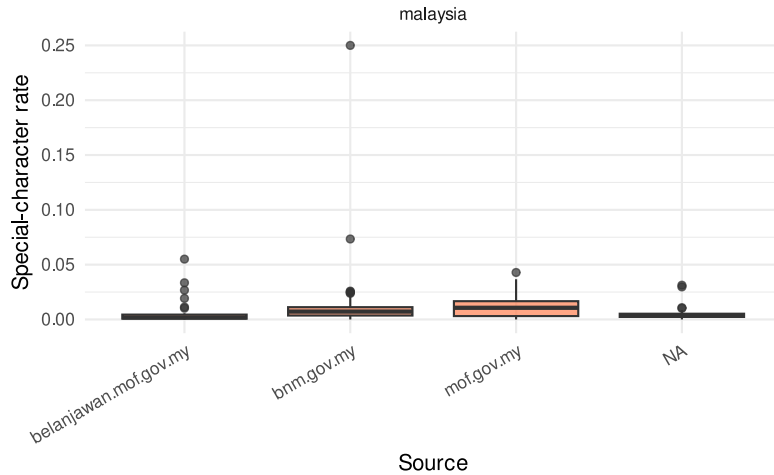


Figure 5: Special-character rate per page, by source

Suspicious-page rate captures OCR garbage (low character count, dense special characters, runaway non-ASCII). Fiscal-vocabulary rate captures whether the extraction preserved the substance of the document. A FAIL here usually means OCR settings need adjustment for some subset of the corpus.

Since the introduction of per-page OCR rescue (image-bearing pages with near-zero native text are routed through Tesseract individually), the dominant residual driver of suspicious pages is legitimately-short cover and section-divider pages with no embedded image — there is nothing for OCR to recover from those. The [OCR pages](#) column in the corpus snapshot tracks the rescue volume per country.

Test (v): Anomaly detection

Diagnostic-only test. Surfaces documents that are unusually short, unusually long, suspected duplicates, or that mark large year-on-year page drops within a series.

Table 9: Test (v) Anomaly counts per country

Country	Short ($\leq 10p$)	Long ($\geq 1000p$)	Duplicates	YoY drops	Status
malaysia	0	0	2	5	INFO

Table 10: Test (v) Flagged anomalous documents

No short or long documents flagged.

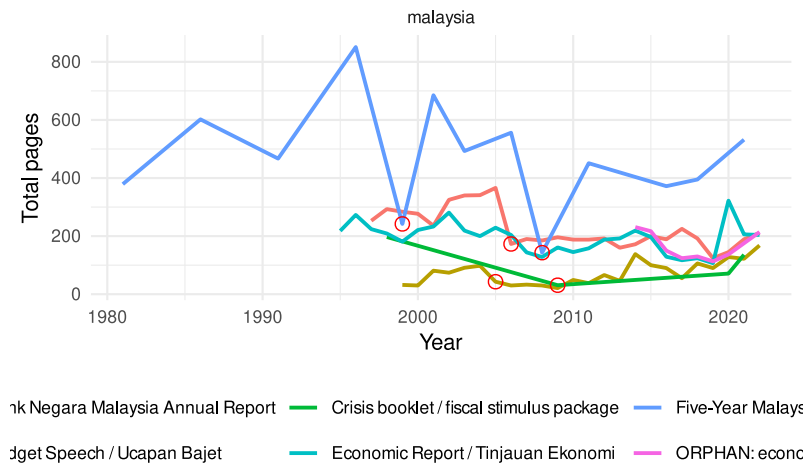


Figure 6: Pages per year and body; red circles mark sudden drops ($>50\%$ YoY)

Anomalies in deployment corpora often have benign explanations (Malaysia’s crisis booklets are intentionally short; bilingual editions may produce duplicate front-matter hashes). The point of this test is awareness, not failure.

Test (vi): Chunk integrity

The US `verify_body.qmd` stops at body-level checks, but the deployment pipeline feeds C1 from `country_chunks`. This test verifies the chunk layer.

Table 11: Test (vi) Chunk-integrity per country

Coun-try	Doc-uments	Pass-ing	Bad start	Bad end	Gap in IDs	Short chunks	Over-sized	Bad markers	Status
malaysia	107	107	0	0	0	0	0	0	PASS

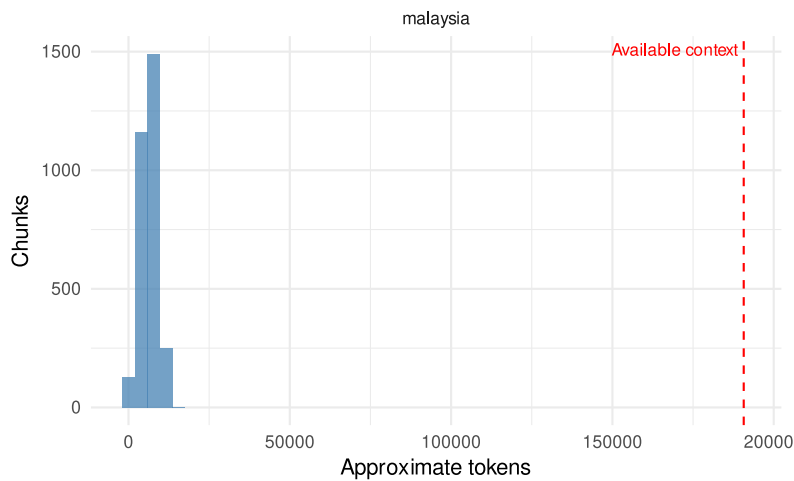


Figure 7: Chunk token distribution with available-context reference line

Any FAIL here is a pipeline bug, not a corpus problem. The most likely failure modes are: a document whose extraction returned zero pages but still produced a chunk row, a marker-count mismatch from a stray `--- PAGE BREAK ---` inside the source text, or an oversized chunk produced by a very long page that the sliding-window logic could not split.

Test (vii): Language audit

`doc_language` claims English for all current Malaysia rows. Bahasa Malaysia contamination is plausible for Hansard fallbacks and bilingual editions. Per chunk we compare a small Bahasa stopword list against an English stopword list and flag chunks where the Bahasa share exceeds the English share.

Table 12: Test (vii) Language audit per country

Country	Chunks sampled	Bahasa-dominant chunks	Bahasa-dominant rate	Status
malaysia	500	138	27.6%	WARN

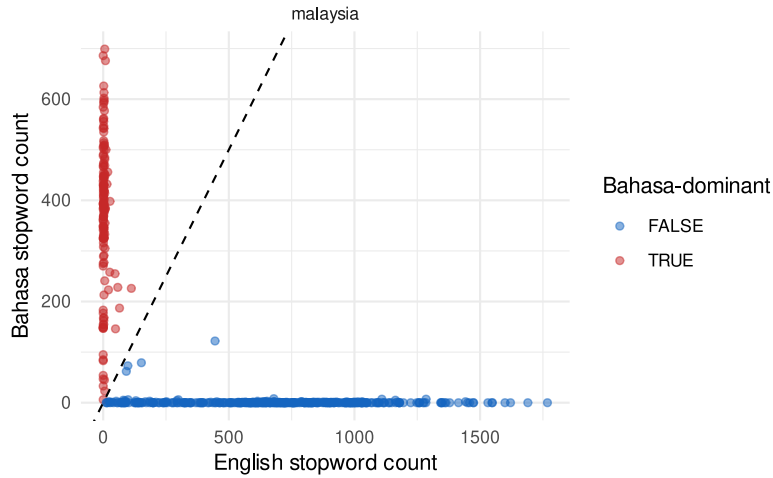


Figure 8: Per-chunk English vs Bahasa stopword density

A small number of Bahasa-dominant chunks is plausible (front matter, signage pages, bilingual sections); a large number suggests a series is misclassified at the `doc_language` field and the codebook prompt’s language assumption may break.

Test (viii): Era stratification of OCR quality

Pre-digital documents are scans and rely on OCR; modern documents are usually born-digital. Quality should be reported separately so a high suspicious-page rate in one era does not drag the corpus-wide average and mask the issue. Era breakpoints are defined per country.

Table 13: Test (viii) OCR usage and quality by era

Country	Era	Documents	OCR rate	Median specials
malaysia	Digital era	104	43.3%	0.005
malaysia	Pre-digital	3	100.0%	—

Pre-digital eras should expect higher OCR usage and higher special-character rates; the values above set a baseline. A future

country whose pre-digital era looks dramatically worse than Malaysia's may need OCR settings tuned before C1 deployment.

Summary dashboard

Table 14: Overall verification status per country

Country	Overall status
malaysia	WARN

Table 15: All verification tests

Country	Test	Metric	Value	Target	Status
malaysia	i	URL resolution rate (excl. pending)	100.0% (99/99)	≥95%	PASS
malaysia	ii	Bound-ary docs ≥10 pages	22/22	All ≥10 pages	PASS
malaysia	iii	Coverage rate (excl. pending)	100.0% (99/99)	≥95%	PASS
malaysia	iv	Suspicious pages	7.6%	≤5%	WARN
malaysia	iv	Fiscal vocab presence	81.2%	≥70%	WARN
malaysia	v	Anomalies detected	0 short / 0 long / 2 dup / 5 YoY drops	Manual review	INFO
malaysia	vi	Chunk-integrity-pass docs	107/107	All pass	PASS
malaysia	vii	Bahasa-dominant chunks (claimed English)	138/500 (27.6%)	0	WARN
malaysia	viii	OCR rate / specials by era	Digital era: OCR 43.3%, specials 0.005 Pre-digital: OCR 100.0%, specials —	Diagnostic	INFO

Next steps

Malaysia — WARN

- Review the following before C1 deployment:
 - Suspicious pages
 - Fiscal vocab presence
 - Bahasa-dominant chunks (claimed English)